

Aditya Kulkarni

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Raleigh, NC -- Open to San Francisco

PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) with production experience building agentic AI systems: Model Context Protocol (MCP) tool execution across 8+ LLMs, Temporal.io durable workflow orchestration, and event-driven multi-step task automation at 2,000 async inference requests/second. At Isomer AI, designed and shipped a unified multi-model agent tooling layer (Claude, GPT-4o, Gemini) with 394 commits and 7 production deployments. Built AlgoPulse to solve the core agentic infrastructure problem -- reliable, fault-tolerant execution of long-running AI reasoning loops at scale. Open to relocating to San Francisco. IEEE-published.

CORE COMPETENCIES

Model Context Protocol (MCP)

Agentic AI Engineering

Temporal.io Durable Workflows

Multi-Model LLM Orchestration

Multi-Step Task Automation

Event-Driven Architecture (Kafka)

Full-Stack Python + TypeScript

Cloud Infrastructure (AWS, GCP, Docker)

WORK EXPERIENCE

Isomer AI May 2025 -- Present

Product Engineer Intern · Remote

- Designed and shipped a **Model Context Protocol (MCP) tool execution layer** for agentic workflows across 3 LLM providers (Anthropic Claude, OpenAI GPT-4o, Google Gemini) -- unified agent tool use protocol for 8+ models in production
- Built **Temporal.io durable workflow orchestration** for multi-step AI task pipelines with Redis session state, real-time streaming (Pusher), and fault-tolerant retry/compensation logic across 7 production deployments
- Shipped 10+ backend features (394 commits) in TypeScript, Node.js, Express, PostgreSQL, Prisma, Docker, Auth0, Sentry -- full-stack ownership from API design to production monitoring
- Implemented document processing agent pipelines: PDF generation, email parsing via Microsoft Graph API, S3 file storage -- multi-step autonomous task execution in production

NC State University April 2025 -- Present

Graduate Research Assistant · Raleigh, NC

- Architected cloud-native genomic data platform on Google Cloud Run -- 500GB datasets, 100+ researchers worldwide; containerized microservice deployment with JBrowse 2
- ML model optimization via hyperparameter tuning and time-series analysis: +12% accuracy improvement with metric-driven deployment decisions

Mphasis (HP Company) Jun 2024 -- Aug 2024

Software Developer Intern · India

- Built 5+ AI-driven automation workflows: OCR/NLP/ML health claim processor (+25% accuracy); AWS and GCP deployments with Grafana observability and production model red-teaming

Capgemini Jun 2023 -- Aug 2023

AI/ML Intern · India

- Built Kafka streaming pipelines for real-time data ingestion (~20% latency) and Hadoop + Spark batch processing (~15% time) -- event-driven architecture for autonomous data workflows at scale

PROJECTS

AlgoPulse -- Agentic AI Platform Independent 2025

Built production **agentic runtime infrastructure** for long-running AI reasoning loops. Identified that API polling fails for agentic workloads and re-architected to **Temporal.io durable execution + Kafka event-driven state synchronization** + Supabase Edge Functions streaming reasoning loops to clients. Result: **2,000 async inference requests/second**, zero state loss under fault conditions.

TypeScript · Node.js · React · Temporal.io · Kafka · Supabase Edge Functions · Redis

Glance AI -- Privacy-First RAG Agent Independent 2025

Engineered a multi-step AI retrieval agent for fintech/healthcare with per-tenant data isolation at the vector retrieval layer. Designed deterministic output constraints preventing hallucinations and cross-tenant context leakage in compliance-sensitive environments.

Python · LLMs · Vector Databases · RAG Architecture

PerCent -- GenAI Finance Agent HackNC24 -- Top 10 / 100+ teams

Built in 48 hours: autonomous financial research agent using RAG + Qwen-14B-Chat via Hugging Face, Selenium-driven web data ingestion, and multi-step task pipeline. Reduced manual research time by 30%. Top 10 of 100+ teams.

Python · Hugging Face · Qwen-14B-Chat · RAG · Selenium

Mapify -- NLP Document Pipeline IEEE Published

T5-transformer pipeline autonomously converting 1,500+ PDFs into structured mind maps -- multi-step document understanding with NLTK and NetworkX. Reduced academic review time 60%. IEEE international conference, 500+ attendees.

T5-Transformer · NLTK · NetworkX · Python

EDUCATION

NC State University -- MS Computer Science Aug 2024 -- May 2026

Raleigh, NC · Coursework: Artificial Intelligence, Algorithms, Data Science, Operating Systems, Software Engineering

Mumbai University -- BTech Computer Science Aug 2020 -- May 2024

CGPA: 9.32/10 · Coursework: Machine Learning, Big Data, Data Structures & Algorithms, OS, Software Engineering

AWARDS & RECOGNITION

HackNC24 Finalist: Top 10 of 100+ teams -- agentic GenAI finance app with RAG and multi-model LLM integration

IEEE Publication: Mapify autonomous NLP pipeline -- presented to 500+ attendees at international conference

National Hackathon Winner: Top 10 in 6+ national hackathons; 3 complete MVPs in 48-hour builds

Google DSC Lead: Led 4+ tech events reaching 500+ students

SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, C, R

Agentic & AI: MCP (Model Context Protocol), OpenAI API, Anthropic Claude, Google Gemini, LangChain, Hugging Face, TensorFlow, PyTorch, RAG, Pandas, NumPy

Orchestration & Streaming: Temporal.io, Kafka, Apache Spark, Hadoop, Redis, event-driven architecture

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Terraform, Grafana, Sentry, CI/CD

Frameworks: Node.js, React, Express, PostgreSQL, Prisma, MERN Stack, Tailwind, Playwright